30 α-particles are observed to be deflected by a thin metal foil through angles that range from 0° to 180°.

Which of the following could explain this observation?

A Due to diffraction from the crystal lattice in the metal.

B Due to reflection from the surface of the foil.

C Due to scattering from electrons in the metal.

D Due to scattering from the small highly positive charged nuclei of atoms in the metal.